

Product Name: VIKANE™**Issue Date:** 12/07/2015

Douglas Products encourages and expects you to read and understand the entire SDS, as there is important information throughout the document. We expect you to follow the precautions identified in this document unless your use conditions would necessitate other appropriate methods or actions.

1. IDENTIFICATION**Product Name:** VIKANE™

Description: End Use Fumigant

US-EPA Reg. No.: 1015-78

COMPANY IDENTIFICATION:**Douglas Products and Packaging Company, LLC**

1550 East Old 210 Highway

Liberty, MO 64068

Customer Information Number: 800-223-3684**EMERGENCY TELEPHONE NUMBER****24-Hour Emergency Contact:** 1-844-845-3129 or 1-352-323-3500**2. HAZARDS IDENTIFICATION****DANGER!**

Fatal if inhaled

Toxic if swallowed



Causes damage to organs (kidney) if inhaled.

May cause damage to organs (respiratory system, nervous system, kidney) through prolonged or repeated exposure if inhaled.

Contains gas under pressure;
may explode if heated.**GHS Toxicity Classifications**

Gases Under Pressure Cat. 1

Acute Toxicity (Inhalation) Cat. 2

Specific Organ Toxicity Cat. 1 (single exposure)

Specific Organ Toxicity Cat. 2 (repeated exposure)

Other Hazards

Persons with pre-existing skin disorders or impaired respiratory or pulmonary function may be at increased risk to the effects of this substance. May cause lung, liver and kidney damage.

Hazard Classification

This material is hazardous under the criteria of the Federal OSHA Hazard Communication Standard 29 CFR 1910.1200.

Prevention

Do not breathe gas. Wash skin thoroughly after handling. Do not eat, drink or smoke when using this product. Use only outdoors or in a well-ventilated area. Wear respiratory protection.

Response

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Immediately call a POISON CENTER or doctor / physician.

IF EXPOSED: Call a POISON CENTER or doctor / physician.

Storage

Store in a well-ventilated place. Keep container tightly closed. Store locked up.

Disposal

Dispose of contents/ container to an approved waste disposal plant.

3. COMPOSITION / INFORMATION ON INGREDIENTS

<u>Common Name</u>	<u>Chemical Name</u>	<u>CASRN</u>	<u>Composition</u>
Sulfuryl fluoride	same	2699-79-8	99.8%

4. FIRST AID MEASURES

General Advice: First Aid responders should pay attention to self-protection and use the recommended protective clothing (chemical resistant gloves, splash protection). If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Most important symptoms and effects, both acute and delayed: Aside from the information found under Description of first aid measures (above) and Indication of immediate medical attention and special treatment needed (below), any additional important symptoms and effects are described in Section 11: Toxicology Information.

IF INHALED: Move person to fresh air. If person is not breathing, call an emergency responder or ambulance, then give artificial respiration; if by mouth to mouth use rescuer protection (pocket mask etc). Call a poison control center or doctor for treatment advice. If breathing is difficult, oxygen should be administered by qualified personnel. If the person is not breathing and has no pulse, consider cardiopulmonary resuscitation (CPR); use pocket resuscitation mask, bag valve mask etc., to avoid risk of poisoning rescuer. To prevent pulmonary edema have the person inhale 5 shots of an aerosol corticosteroid metered dose inhaler (if available), such as beclomethasone or fluticasone, etc., every 10 minutes until the person is evaluated by a physician.

IF ON SKIN OR CLOTHING: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice. In case of frostbite, immediately flush skin with plenty of water for 15 minutes. Seek medical attention. Suitable emergency safety shower facility should be immediately available.

IF IN EYES: In case of frostbite, immediately flush eyes with water; remove contact lenses, if present, after the first 5 minutes, then continue flushing eyes for at least 15 minutes. Obtain medical attention promptly, preferably from an ophthalmologist. Suitable emergency eye wash facility should be immediately available.

IF SWALLOWED: Call a poison control center or doctor immediately for treatment advice. Have person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by the poison control center or doctor. Never give anything by mouth to an unconscious person.

NOTE TO PHYSICIAN: Maintain adequate ventilation and oxygenation of the patient. Sulfuryl fluoride is a gas which has no warning properties such as odor or eye irritation. The prediction of possible human effects is based in part on observations made on laboratory animals. Treat for frostbite if present (eyes, skin) with gentle rewarming by water irrigation for at least 15 minutes. It is predicted that persons exposed to sulfuryl fluoride will show little evidence of intoxication at first, unless the concentration is very high (greater than 400 ppm). Early symptoms of exposure to sulfuryl fluoride are respiratory irritation and central nervous system depression. Excitation may follow. Slowed movement, reduced awareness, and slow or garbled speech may be noted. It is essential to keep such an individual at bed rest for at least 24 hours. Clinical observations should be directed at the pulmonary, hepatic, and renal systems. Prolonged exposure can produce lung irritation, pulmonary edema, nausea, and abdominal pain. Repeated exposure to high concentrations can result in significant lung and kidney damage. Convulsions may ensue with respiratory arrest being the terminal event. Assisted respiration may be necessary. Clinical observation is essential. There is no known antidote for overexposure to sulfuryl fluoride. May cause asthma-like (reactive airways) symptoms. Bronchodilators, expectorants, antitussives and corticosteroids may be of help. Respiratory symptoms, including pulmonary edema, may be delayed. Persons receiving significant exposure should be observed 24-48 hours for signs of respiratory distress. Consider administering a complete aerosol corticosteroid metered dose inhaler (100-150 shots) or equivalent as initial preventive treatment for incipient pulmonary edema. Consider administering 250-1000 mg prednisolone IV on the first day of treatment. Treat for frostbite, if present. No specific antidote. Treatment of exposure should be directed at the control of symptoms and the clinical condition of the patient. Have the Safety Data Sheet, and if available, the label with you when calling a poison control center or doctor, or going for treatment. Excessive exposure may aggravate preexisting asthma and other respiratory disorders (e.g. emphysema, bronchitis, reactive airways dysfunction syndrome).

5. FIREFIGHTING MEASURES

Suitable extinguishing media: This material does not burn. If exposed to fire from another source, use suitable extinguishing agent for that fire.

Unsuitable extinguishing media: No data available.

Hazardous combustion products: Fire conditions may cause this product to decompose. Refer to Section 10 - Thermal Decomposition.

Unusual Fire and Explosion Hazards: Container may rupture from gas generation in a fire situation.

Fire Fighting Procedures: Keep people away. Isolate fire and deny unnecessary entry. Stay upwind. Keep out of low areas where gases (fumes) can accumulate. Use water spray to cool fire exposed containers and fire affected zone until fire is out and danger of reignition has passed. Fight fire from protected location or safe distance. Consider the use of unmanned hose holders or monitor nozzles. Immediately withdraw all personnel from the area in case of rising sound from venting safety device or discoloration of the container. Move container from fire area if this is possible without hazard. Contain fire water run-off if possible. Fire water run-off, if not contained, may cause environmental damage. Review the "Accidental Release Measures" and the "Ecological Information" sections of this SDS.

Special protective equipment for firefighters: Wear positive-pressure self-contained breathing apparatus (SCBA) and protective firefighting clothing (includes firefighting helmet, coat, trousers, boots, and gloves). Avoid contact with this material during firefighting operations. If contact is likely, change to full chemical resistant firefighting clothing with self-contained breathing apparatus. If this is not available, wear full chemical resistant clothing with self-contained breathing apparatus and fight fire from a remote location. For protective equipment in post-fire or non-fire clean-up situations, refer to the relevant sections.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures: Isolate area. Keep personnel out of low areas. Keep upwind of spill. Ventilate area of leak or spill. Use appropriate safety equipment. For additional information, refer to Section 8, Exposure Controls and Personal Protection.

Environmental precautions: Prevent from entering into soil, ditches, sewers, waterways and/or groundwater. See Section 12, Ecological Information.

Methods and materials for containment and cleaning up: *Isolate area until gas has dispersed.*

Small spills: Knock down and dilute vapors with water fog or spray. Apply vapor suppression foams until spill can be cleaned up. Use non-sparking tools in cleanup operations.

Large spills: Contact Douglas Products for clean-up assistance. See Section 13, Disposal Considerations, for additional information.

7. HANDLING AND STORAGE

Precautions for safe handling: Keep out of reach of children. Avoid contact with eyes, skin, and clothing. Do not swallow. Avoid breathing vapor. Wash thoroughly after handling. Keep container closed. Use with adequate ventilation. See Section 8, EXPOSURE CONTROLS AND PERSONAL PROTECTION.

Conditions for safe storage: Store in a dry place. Store in original container. Keep container tightly closed when not in use. Do not store near food, foodstuffs, drugs or potable water supplies

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Control Parameters

Exposure limits are listed below, if they exist.

Component	Regulation	Type of Listing	Value / Notation
Sulfuryl fluoride	ACGIH	TWA	5 ppm
		STEL	10 ppm
	OSHA Z-1 ACGIH	TWA	5 ppm, 20 mg/m ³
		TWA	BEI
		STEL	BEI

RECOMMENDATIONS IN THIS SECTION ARE FOR MANUFACTURING, COMMERCIAL BLENDING AND PACKAGING WORKERS. APPLICATORS AND HANDLERS SHOULD SEE THE PRODUCT LABEL FOR PROPER PERSONAL PROTECTIVE EQUIPMENT AND CLOTHING.

Exposure Controls

Engineering Controls: Use engineering controls to maintain airborne level below exposure limit requirements or guidelines. If there are no applicable exposure limit requirements or guidelines, use only in enclosed systems or with local exhaust ventilation. Exhaust systems should be designed to move the air away from the source of vapor/aerosol generation and people working at this point. Lethal concentrations may exist in areas with poor ventilation.

Individual Protection Measures

Eye/face Protection: For handling the gas, wear safety glasses (with side shields). When contact with the liquid (condensed gas) is possible, wear chemical goggles.

Skin Protection: Wear clean, body-covering clothing. Chemical protective gloves should not be needed when handling this material. Consistent with general hygienic practice for any material, skin contact should be minimized.

Respiratory Protection: Respiratory protection should be worn when there is a potential to exceed the exposure limit requirements or guidelines. If there are no applicable exposure limit requirements or guidelines, use an approved respirator. When respiratory protection is required, use an approved positive-pressure self-contained breathing apparatus or positive-pressure airline with auxiliary self-contained air supply. For emergency conditions, use an approved positive-pressure self-contained breathing apparatus. In confined or poorly ventilated areas, use an approved self-contained breathing apparatus or positive pressure air line with auxiliary self-contained air supply.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical State	Gas
Color	Colorless
Odor	Odorless
Odor Threshold	Odorless
pH	Not applicable
Melting Point	-137 °C (-215 °F) Estimated
Freezing Point	Not applicable
Boiling point (EC Method A2)	-54 °C (-65 °F)
Flash point (closed cup)	Not applicable
Evaporation Rate	Not applicable
Flammability (solid, gas)	Not Flammable
Lower Explosion Limit	Not applicable
Upper Explosion Limit	Not applicable
Vapor Pressure	18,000 hPa at 20 °C (68 °F)
Relative Vapor Density (air = 1)	3.5 at 20 °C (68 °F)
Relative Density (water = 1)	1.35
Water Solubility	1.04 g/L 20°C, Unbuffered
Auto-ignition Temperature	Not applicable
Decomposition Temperature	no data available
Kinematic Viscosity	no data available
Explosive Properties	no data available
Oxidizing Properties	no data available
Molecular Weight	no data available

NOTE: The physical data presented above are typical values and should not be construed as a specification.

10. STABILITY AND REACTIVITY

Reactivity: No dangerous reaction known under conditions of normal use.

Chemical stability: Stable under recommended temperatures and pressures.

Possibility of hazardous reactions: Hazardous polymerization will not occur.

Conditions to avoid: Exposure to elevated temperatures can cause product to decompose. Generation of gas during decomposition can cause pressure in closed systems.

Incompatible materials: Strong bases.

Hazardous decomposition products: Decomposition products can include and are not limited to: Hydrogen fluoride. Sulfur oxides. Toxic gases are released during decomposition.

11. TOXICOLOGICAL INFORMATION

Toxicological information appears in this section when such data is available.

Acute Oral Toxicity

Moderate toxicity if swallowed. Swallowing is unlikely because of the physical state. Single dose oral LD₅₀ has not been determined.

Acute Dermal Toxicity

Prolonged skin contact is unlikely to result in absorption of harmful amounts. The dermal LD₅₀ has not been determined.

Acute Inhalation Toxicity

Vapor concentrations are attainable which may be fatal with single exposure. Excessive exposure may cause severe irritation to upper respiratory tract (nose and throat) and lungs. For narcotic effects: Relevant data not available.

LC₅₀, Rat, 4 Hour, gas, 991 - 1122 ppm

LC₅₀, Mouse, 4 Hour, gas, 400 - 600 ppm

Skin Corrosion / Irritation

Essentially nonirritating to skin. Liquid may cause frostbite upon skin contact.

Serious Eye Damage / Eye Irritation

No hazard from gas. Liquid may cause frostbite.

Sensitization

No relevant data found.

Specific Target Organ Systemic Toxicity (Single Exposure)

Causes damage to organs.

Route of Exposure: Inhalation

Target Organs: Kidney

Specific Target Organ Systemic Toxicity (Repeated Exposure)

In animals, effects have been reported on the following organs:

Central nervous system, Kidney, Lung, Respiratory tract, Thyroid.

Observations in animals include: Convulsions, Tremors.

May cause fluorosis of teeth and bones.

Carcinogenicity

Did not cause cancer in laboratory animals.

Teratogenicity

Has been toxic to the fetus in laboratory animals at doses toxic to the mother. Did not cause birth defects in laboratory animals.

Reproductive Toxicity

In animal studies, did not interfere with reproduction.

Mutagenicity

Most in vitro genetic toxicity studies were negative, but some were positive due to artifacts associated with the test system.. Animal genetic toxicity studies were negative.

Aspiration Hazard

Based on physical properties, not likely to be an aspiration hazard.

12. ECOLOGICAL INFORMATION

Ecotoxicological information appears in this section when such data is available.

Acute toxicity to fish

LC₅₀, *Danio rerio* (zebra fish), static test, 96 Hour, 0.89 mg/l

Acute toxicity to aquatic invertebrates

EC₅₀, *Daphnia magna* (Water flea), static test, 48 Hour, 0.62 mg/l

Acute toxicity to algae/aquatic plants

EyC₅₀, *Pseudokirchneriella subcapitata* (green algae), static test, 96 Hour, Growth inhibition (cell density reduction), 3.05 mg/l

EbC₅₀, *Pseudokirchneriella subcapitata* (green algae), static test, 72 Hour, Biomass, 0.58 mg/l

ErC₅₀, *Pseudokirchneriella subcapitata* (green algae), static test, 72 Hour, Growth rate inhibition, 1.13 mg/l

Toxicity to Above Ground Organisms

LC₅₀, *Apis mellifera* (bees), 2 Hour, mortality, 6.5mg/l

LC₅₀, *Colinus virginianus* (Bobwhite quail), 4 Hour, 1,844 ppm

Persistence and degradability

Biodegradability: Chemical degradation (hydrolysis) is expected in the environment.

Bioaccumulative potential

Bioaccumulation: Bioconcentration potential is low (BCF < 100 or Log Pow < 3).

Partition coefficient: n-octanol/water (log Pow): 0.41 Estimated.

Mobility in soil

Potential for mobility in soil is very high (Koc between 0 and 50).

Partition coefficient (Koc): 6 Estimated.

13. DISPOSAL CONSIDERATIONS

Disposal Methods: If wastes and/or containers cannot be disposed of according to the product label directions, disposal of this material must be in accordance with your local or area regulatory authorities. This information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations. If the material as supplied becomes a waste, follow all applicable regional, national and local laws.

14. TRANSPORT INFORMATION**Classification for LAND Transport (DOT):**

Proper Shipping Name: UN2191 Sulfuryl Fluoride, 2.3

Classification for SEA Transport (IMO-IMDG):

Proper Shipping Name: UN2191 Sulfuryl Fluoride, 2.3, Marine Pollutant

Classification for AIR Transport (IATA/ICAO):

Transportation by air prohibited.

This information is not intended to convey all specific regulatory or operational requirements/information relating to this product. Transportation classifications may vary by container volume and may be influenced by regional or country variations in regulations. Additional transportation system information can be obtained through an authorized sales or customer service representative. It is the responsibility of the transporting organization to follow all applicable laws, regulations and rules relating to the transportation of the material.

15. REGULATORY INFORMATION**Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)**

EPA Registration Number: 1015-78

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals.

Following is the hazard information as required on the pesticide label:

DANGER

Extremely hazardous liquid and vapor under pressure

Fatal if inhaled

May be fatal if swallowed

Liquid causes freeze burns on exposed skin

OSHA Hazard Communication Standard

This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.

Superfund Amendments and Reauthorization Act of 1986 Title III (Emergency Planning and Community Right-to-Know Act of 1986) Sections 311 and 312

Immediate (Acute) Health Hazard	Yes
Delayed (Chronic) Health Hazard	Yes
Fire Hazard	No
Reactive Hazard	No
Sudden Release of Pressure Hazard	Yes

Superfund Amendments and Reauthorization Act of 1986 Title III (Emergency Planning and Community Right-to-Know Act of 1986) Section 313

This product contains the following substances which are subject to the reporting requirements of Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 and which are listed in 40 CFR 372.

<u>Component</u>	<u>CAS #</u>	<u>Amount</u>
Sulfuryl fluoride	2699-79-8	99.8%

Pennsylvania Worker and Community Right-To-Know Act:

Sulfuryl Fluoride is cited in the Pennsylvania Hazardous Substance List at levels which require reporting. A component of this product (1,2-Dichloroethane) is cited in the Pennsylvania Special Hazardous Substance List at levels which require reporting.

California Proposition 65 (Safe Drinking Water and Toxic Enforcement Act of 1986)

This product contains no listed substances known to the State of California to cause cancer, birth defects or other reproductive harm, at levels which would require a warning under the statute.

United States TSCA Inventory (TSCA)

This product contains chemical substance(s) exempt from U.S. EPA TSCA Inventory requirements. It is regulated as a pesticide subject to Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA) requirements.

16. OTHER INFORMATION

NFPA Rating:

Health Hazard: 4 Fire Hazard: 0 Reactivity Hazard: 0

Douglas Products and Packaging Company, LLC urges each customer or recipient of this SDS to study it carefully and consult appropriate expertise, as necessary or appropriate, to become aware of and understand the data contained in this SDS and any hazards associated with the product. The information herein is provided in good faith and believed to be accurate as of the effective date shown above. However, no warranty, express or implied, is given. Regulatory requirements are subject to change and may differ between various locations. It is the buyer's/user's responsibility to ensure that his activities comply with all federal, state, provincial or local laws. The information presented here pertains only to the product as shipped. Since conditions for use of the product are not under the control of the manufacturer, it is the buyer's/user's duty to determine the conditions necessary for the safe use of this product. Due to the proliferation of sources for information such as manufacturer-specific SDSs, we are not and cannot be responsible for SDSs obtained from any source other than ourselves. If you have obtained an SDS from another source or if you are not sure that the SDS you have is current, please contact us for the most current version.

SDS Version: 1.0 (12/07/2015)

